

Carnegie Mellon University

# Case Study: LendingClub Prescriptive Analytics

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# Optimized Loan Portfolio Construction

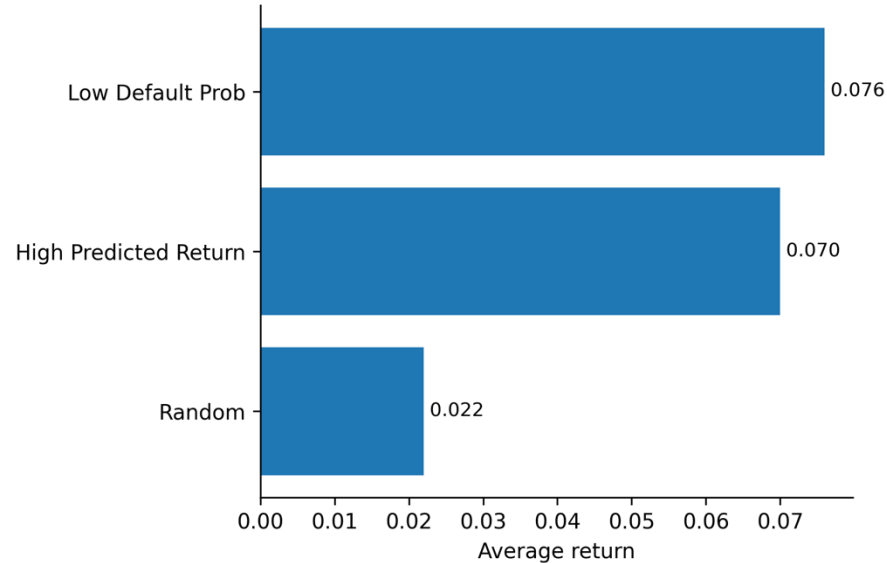
with Lending Constraints

- Build a prescriptive optimization model to outperform simple heuristics
- LendingClub hold-out pool
  - Single-grade “C” loans, sparse rates
- Key metrics
  - Expected return ( $\hat{\mu}$ ), risk ( $\hat{\sigma}$ ), cost, grade caps

# Model Framework

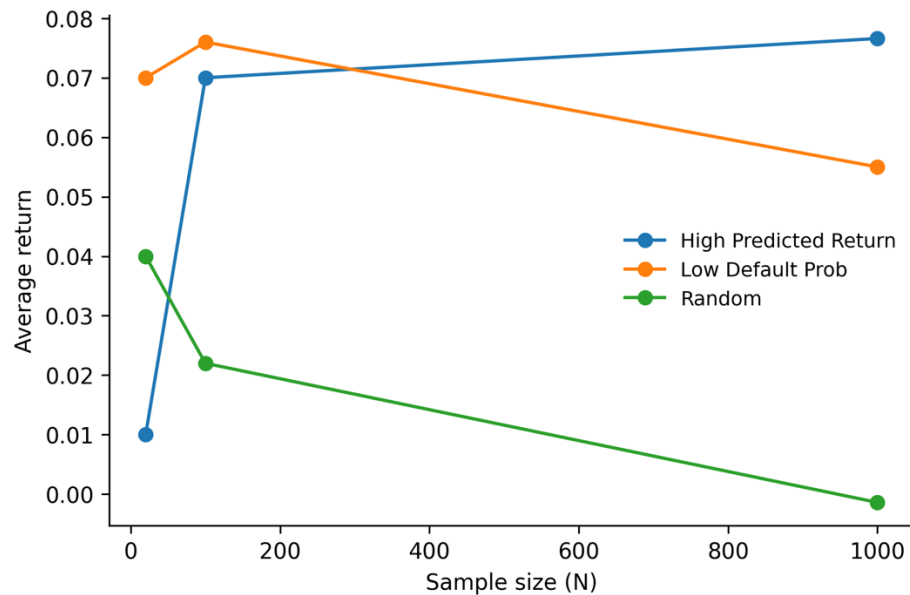
- Baseline Heuristics
  - Top- $\mu$ , Top-p(ok), Two-Stage, Greedy  $\mu/\sigma$
- Prescriptive Models
  - LP: maximize  $\sum \hat{\mu}_i x_i$  s.t. budget + risk + grade caps
  - QP: mean–variance trade-off  $\max \sum \hat{\mu}_i x_i - \lambda x^T \Sigma x$
- Constraints
  - Budget  $\leq B$ ,  $n \leq N_{\max}$
  - $\sigma \leq R_{\max}$ , grade share  $\leq 35\%$

# Investment Strategy Comparison



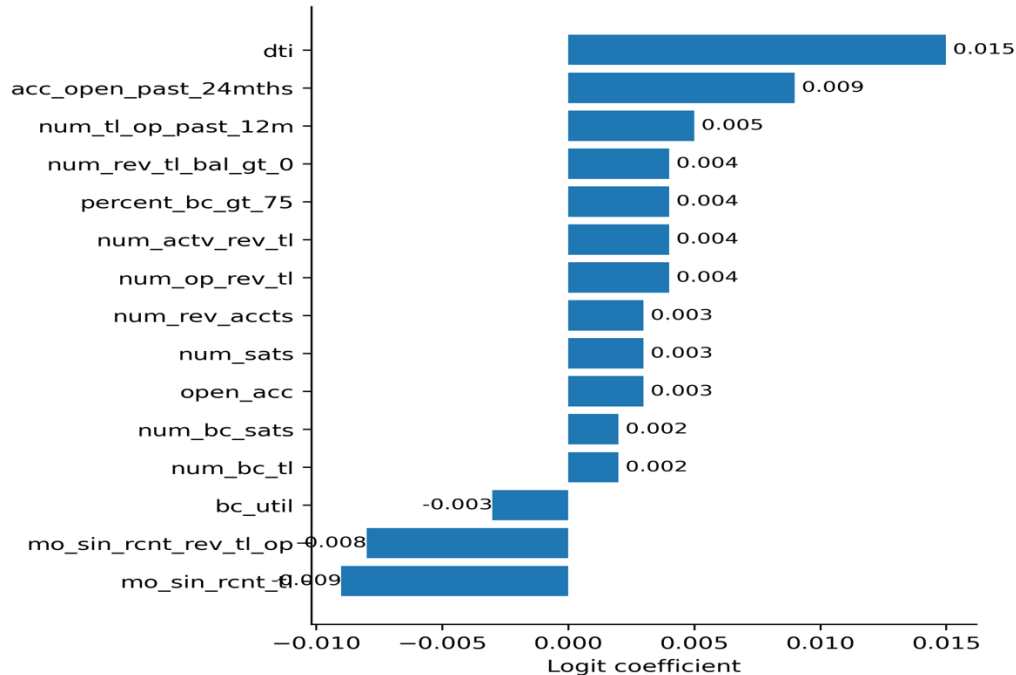
**Figure 01.** Low Default Probability portfolio achieves the highest realized average return (7.6%) versus High Predicted Return (7.0%) and Random (2.2%).

# Portfolio Stability & Sensitivity Analysis



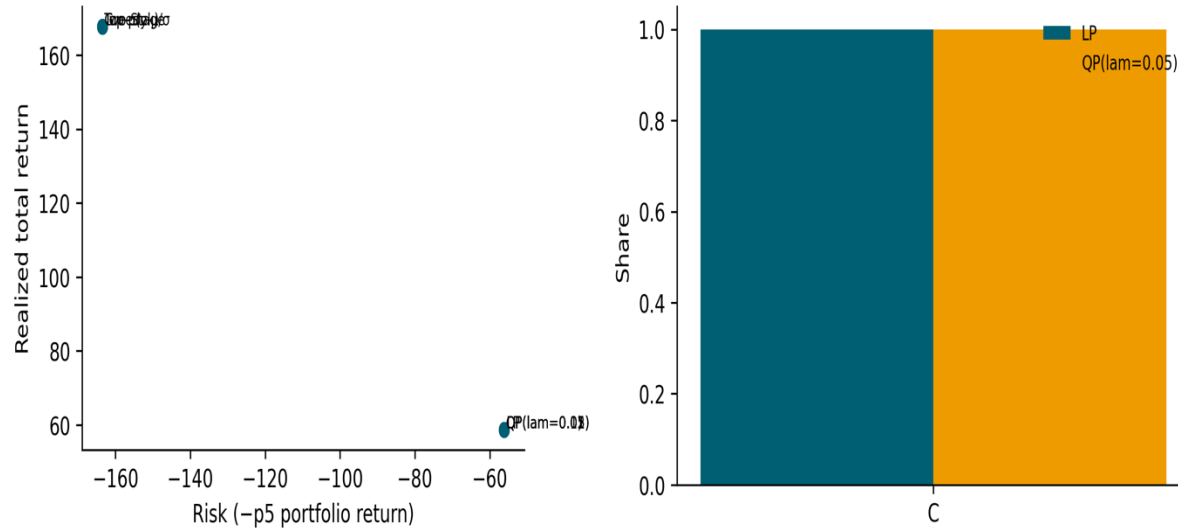
**Figure 01.** Performance stability across increasing sample sizes ( $N = 20 \rightarrow 1000$ ). High Predicted Return scales efficiently; Low Default Probability remains stable.

# Investment Strategy Comparison



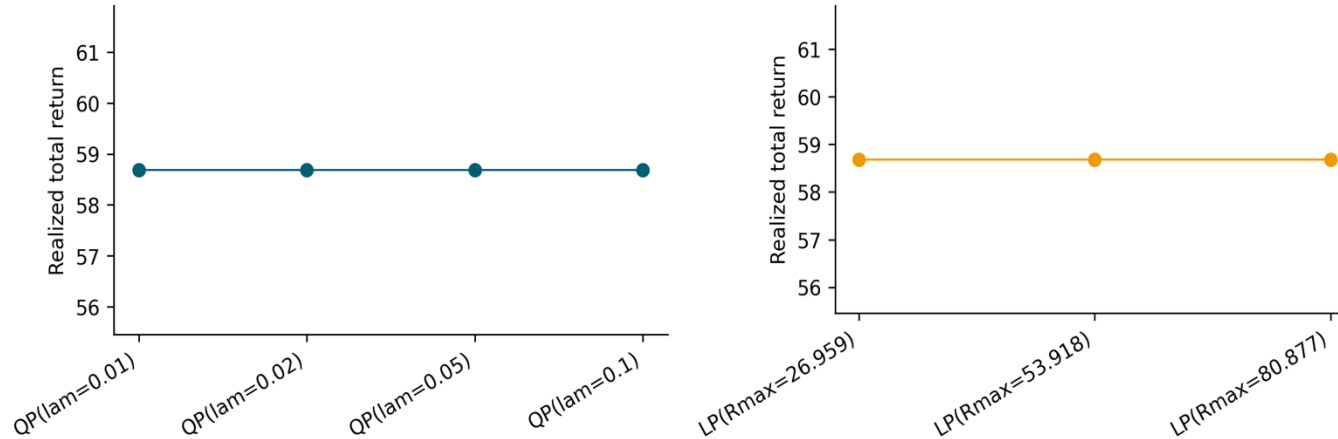
**Figure 01.** Top 15 logistic regression coefficients influencing predicted default probability.

# Risk-Return Frontier & Grade Diversification



**Figure 01.** LP and QP lie on the same efficient point due to the grade cap; both maintain 100% grade C exposure but control risk through selection count.

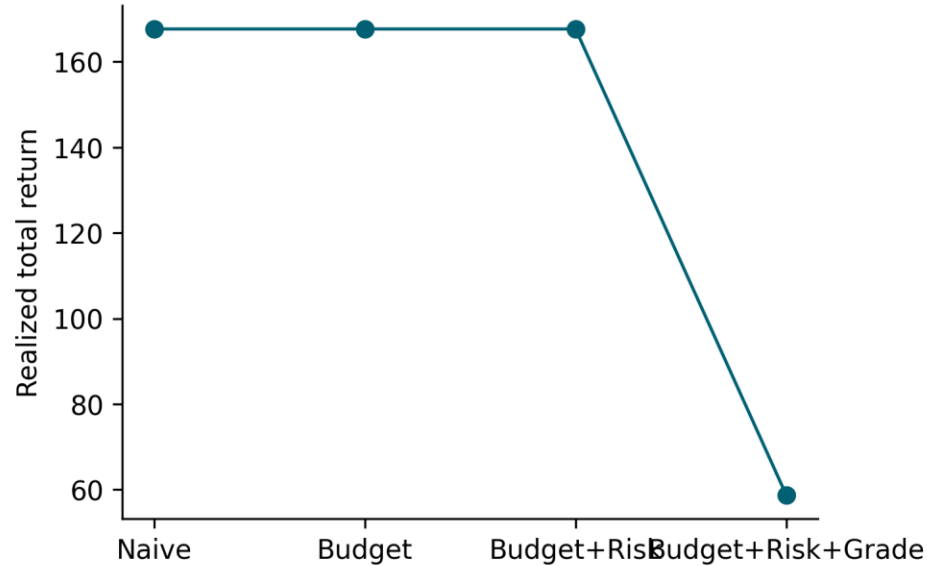
# Policy Sensitivity Impact



**Figure 02.** Signals depict higher DTI/lower income/certain purposes correlated with increased risk; home ownership/tenure increases in correlation with stability.



# Policy Constraint Impact



**Figure 02.** Signals depict higher DTI/lower income/certain purposes correlated with increased risk; home ownership/tenure increases in correlation with stability.

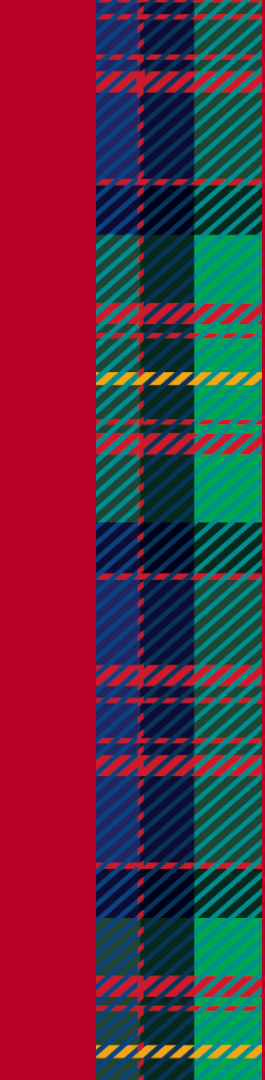
# Recommendation

## Immediate Action

- Deploy LP portfolio
  - Same performance as QP
- Prioritize loans with the lowest predicted default probability
  - Maximizes stable returns
- Maintain 35% grade cap for diversification control
- Re-run quarterly as new grades enter pipeline
- Include proper figures running on clean data
- Risk of possible over-conservatism limits in large sample portfolios
  - Combine with High Predicted Return weighting in larger datasets

# References

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Thank You!

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